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Optimal estimation of linear operators from inaccurate data: A second look

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This paper intends to re-examine some results and proofs given in a previous publication on optimal estimation under uncertainty. In a rather general setting we showed that regularization of an element of a linear space relative to a quadratic criterion and inaccurate linear observations is an optimal method for recovering a linear operator of that element. For this to be the case, the regularization parameter must be chosen with care.

1. Introduction

This paper intends to re-examine some results and proofs given in a previous publication on optimal estimation under uncertainty as well as draw some further conclusions about this subject. In a rather general setting we showed in [5] that regularization of an element of a linear space relative to a quadratic criterion and inaccurate linear observations is an optimal method for recovering a linear operator of that element among *all* algorithms, even ones that are nonlinear in the observation. For this to be the case, the regularization parameter must be chosen with care.

An essential ingredient in the proof was a geometric observation about the set

$$\mathcal{M} = \{(\|x\|_1^2, \|x\|_2^2) : \|x\|_0 = 1,\ x \in \mathcal{H}\}, \tag{1.1}$$

where $\|x\|_i$, $i = 0, 1, 2$, are semi-definite quadratic forms on a linear space $\mathcal{H}$. Specifically, we showed in [5] that $\mathcal{M}$ has a convex profile, that is, the boundary of the convex hull of $\mathcal{M}$ is contained in $\mathcal{M}$.

At the time of writing [5] we were not aware of a result of Brickman [1] which says, if $(\mathcal{H}, \|\bullet\|_0)$ is a *finite* dimensional Hilbert space with dimension at least three, then $\mathcal{M}$ is in fact *convex*. We first became aware of this result in conversation with our colleague Alan Hoffman of IBM, to whom we are grateful. Brickman's work sparked our renewed interest in the material in [5]. Adding to this the several recent papers [2,4,6,11], which either use the results in [5] or would benefit by their application, we have been encouraged to write this paper.

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In section 2, we prove that $\mathcal{M}$ is convex if $\dim\mathcal{H} \geqslant 3$, even when $\mathcal{H}$ is infinite dimensional. The proof is similar to that of [1], but differs in some details which allow us to avoid requiring that $\| \bullet \|_0$ is a quadratic *norm* and $\mathcal{H}$ is *finite* dimensional. In section 3, we apply this result to optimal recovery in Hilbert spaces and in section 4 identify the form of the optimal algorithm for recovering a function from a finite number of its moments, a question left unsettled in [10]. For recent work on this problem see [6–8]. The last section contains some observation about optimal information operators.

2. Field of values in a Hilbert space

In this section we provide a proof of the fact that given any linear space $\mathcal{H}$ of dimension at least three and any three quadratic semi-norms $\| \bullet \|_i$, $i = 0, 1, 2$ on $\mathcal{H}$, the set

$$\mathcal{M} = \{(\|x\|_1^2, \|x\|_2^2) : \|x\|_0 = 1,\ x \in \mathcal{H}\} \tag{2.1}$$

is a convex subset of $\mathbb{R}_+^2$. This result in the finite dimensional case is due to Brickman [1]. The proof we present below differs in detail but not in spirit from his proof.

PROPOSITION 2.1

If $\dim\mathcal{H} \geqslant 3$ and $\| \bullet \|_0$ is a norm then $\mathcal{M}$ is convex.

Proof

Let

$$\begin{aligned} a^0 &= (\|x^0\|_1^2, \|x^0\|_2^2), \quad \|x^0\|_0^2 = 1\,, \\ a^1 &= (\|x^1\|_1^2, \|x^1\|_2^2), \quad \|x^1\|_0^2 = 1\,, \end{aligned} \tag{2.2}$$

be two distinct points in $\mathcal{M}$. Suppose for some α, β, γ with $\alpha^2 + \beta^2 + \gamma^2 > 0$,

$$\alpha u + \beta v + \gamma = 0, \quad (u, v) \in \mathbb{R}^2\,, \tag{2.3}$$

is an equation of a line containing a^0 and a^1. To express this condition in a convenient form we introduce the symmetric indefinite bi-linear form

$$\langle x, y\rangle := \alpha(x, y)_1 + \beta(x, y)_2 + \gamma(x, y)_0 \tag{2.4}$$

so that $\langle x^0, x^0\rangle = \langle x^1, x^1\rangle = 0$. We will construct a continuous path $x(t) : [0, 1] \to \mathcal{H}$ such that $\langle x(t), x(t)\rangle = 0, \|x(t)\|_0^2 \neq 0, 0 \leqslant t \leqslant 1$, and either $x(0) = x^0, x(1) = x^1$ or $x(0) = -x^0, x(1) = x^1$. In either case the path

$$V(t) := (\|x(t)\|_1^2/\|x(t)\|_0^2, \|x(t)\|_2^2/\|x(t)\|_0^2), \quad t \in [0, 1]\,, \tag{2.5}$$

lies within $\mathcal{M}$ and traverses the line segment joining a^0 to a^1. Hence, every point on that line segment is in $\mathcal{M}$.

To construct the path we consider three distinct cases:

Case 1. Here we assume that $\langle x^0, x^1\rangle = 0$ and choose

$$x(t) = (1-t)x^0 + tx^1 . \tag{2.6}$$

If $\|x(t)\|_0 = 0$ for some $t \in (0,1)$ then it follows that x^0 and x^1 are linearly dependent. Thus a^0 and a^1 must be the same vector. This contradiction insures that $\|x(t)\|_0 \neq 0$ for $t \in [0,1]$ and so (2.6) will do.

Next, we consider what happens when $\langle x^0, x^1\rangle \neq 0$. We divide this possibility into two cases.

Case 2. Suppose there is a $u \in \mathcal{H}$ such that $\langle x^0, u\rangle = \langle x^1, u\rangle = 0$ while $\langle u, u\rangle \neq 0$ and consider the path

$$x(t) = \begin{cases} (1-t)^2 x^0 + rt(1-t)u + t^2 x^1, & \text{if } \kappa := \langle x^0, x^1\rangle\langle u, u\rangle < 0, \\ -(1-t)^2 x^0 + rt(1-t)u + t^2 x^1, & \text{if } \kappa > 0, \end{cases} \tag{2.7}$$

where

$$r = \sqrt{2\left|\frac{\langle x^0, x^1\rangle}{\langle u, u\rangle}\right|} .$$

When $\kappa < 0$, $x(t)$ connects x^0 to x^1 while in the other case it connects $-x^0$ to x^1. Geometrically speaking, the hyperconic $\langle x, x\rangle = 0$ has two pieces and either x^0, x^1 or $-x^0, x^1$ are in the same piece. Now, in either case we have

$$\begin{aligned} \langle x(t), x(t)\rangle &= r^2 t^2 (1-t)^2 \langle u, u\rangle - 2 \operatorname{sgn} \kappa t^2 (1-t)^2 \langle x^0, x^1\rangle \\ &= t^2 (1-t)^2 \{r^2 \langle u, u\rangle - 2 \operatorname{sgn} \kappa < x^0, x^1\rangle\} \\ &= 0 . \end{aligned}$$

Moreover, since

$$\langle x(t), u\rangle = rt(1-t)\langle u, u\rangle ,$$

we see that $x(t) \neq 0$, for $t \in [0,1]$ and therefore we again have $\|x(t)\|_0 \neq 0$ for $t \in [0,1]$.

Case 3. The final case we need to consider occurs whenever $u \in \mathcal{H}$ and $\langle x^0, u\rangle = \langle x^1, u\rangle = 0$ it follows that $\langle u, u\rangle = 0$. In this case, we claim that for all $z \in \mathcal{H}$

$$\langle z, z\rangle = 2\frac{\langle z, x^0\rangle\langle z, x^1\rangle}{\langle x^0, x^1\rangle} . \tag{2.8}$$

For the proof of this formula we consider the vector

$$u = \langle z, x^1\rangle x^0 + \langle z, x^0\rangle x^1 - z\langle x^0, x^1\rangle,$$

which was chosen so that $\langle x^0, u\rangle = \langle x^1, u\rangle = 0$. Hence

$$\begin{aligned} 0 = \langle u, u\rangle &= \langle z, z\rangle\langle x^0, x^1\rangle^2 + 2\langle z, x^1\rangle\langle z, x^0\rangle\langle x^0, x^1\rangle \\ &\quad - 2\langle z, x^0\rangle\langle z, x^1\rangle\langle x^0, x^1\rangle - 2\langle z, x^0\rangle\langle z, x^1\rangle\langle x^0, x^1\rangle \\ &= \langle x^0, x^1\rangle\{\langle z, z\rangle\langle x^0, x^1\rangle - 2\langle z, x^0\rangle\langle z, x^1\rangle\}, \end{aligned}$$

which proves (2.8).

To construct a path connecting x^0 and x^1 we choose any $z^0 \in \mathcal{H}/\{0\}$ which satisfies

$$\langle x^0, z^0\rangle = \langle z^1, z^0\rangle = 0.$$

Such a z^0 exists since $\dim \mathcal{H} \geqslant 3$. Moreover, by (2.8) $\langle z^0, z^0\rangle = 0$ and z^0 is not a multiple of either x^0 or x^1. If

$$x(t) = \begin{cases} (1-2t)x^0 + 2tz^0, & 0 \leqslant t \leqslant \frac{1}{2} \\ 2(1-t)z^0 + (2t-1)x^1, & \frac{1}{2} \leqslant t \leqslant 1 \end{cases}, \tag{2.9}$$

then $\langle x(t), x(t)\rangle = 0$ and also $x(t) \neq 0$ for $t \in [0, 1]$. □

Brickman observed when $\dim \mathcal{H} < \infty$ the set $\mathcal{M}$ is the numerical range of the matrix $A_1 + iA_2$, where

$$(x, x)_i = (A_i x, x)_0, \quad i = 1, 2.$$

Consequently, several useful facts about $\mathcal{M}$, especially the form of its bounding curves, follow from results in Kippenhahn [3].

In applications it is essential to relax the hypothesis of proposition 2.1 that $\|\bullet\|_0$ is a norm. If instead $\|\bullet\|_0$ is only a semi-norm then the set

$$\mathcal{N} = \{x : x \in \mathcal{H}, \|x\|_0 = 0\}$$

forms a nontrivial subspace in $\mathcal{H}$. In this case, the hypothesis that $\dim \mathcal{H} \geqslant 3$ is interpreted as follows. There does not exist a subspace $\mathcal{S} \subseteq \mathcal{H}$, with $\dim \mathcal{S} \leqslant 2$ such that

$$\mathcal{H} = \mathcal{S} + \mathcal{N},$$

equivalently, $\dim \mathcal{H}/\mathcal{N} \geqslant 3$. Let us say, for simplicity even in this case that $\dim \mathcal{H} \geqslant 3$. Then we have

THEOREM 2.1

Let $\mathcal{H}$ be a linear space and $\|x\|_i$, $i = 0, 1, 2$, be quadratic semi-norms defined on $\mathcal{H}$. If $\dim \mathcal{H} \geqslant 3$ then $\mathcal{M}$ is convex.

Proof

We organize the proof just as in the proof of proposition 2.1. In the case at hand the path $x(t) : [0,1] \to \mathcal{H}$ with $\langle x(t), x(t)\rangle = 0, t \in [0,1]$ that we will construct connects a vector $x(0) = \tilde{x}^0$ to $x(1) = \tilde{x}^1$ where each $\tilde{x}^i$ is some (nonzero) multiple of x^i. Therefore, (2.5) will still traverse the line segment joining a^0 to a^1 and lie in $\mathcal{M}$.

Case 1. First we suppose $\langle x^0, x^1\rangle_0 = 0$. For (2.6) if $\|x(t)\|_0 = 0$ for some $t \in (0,1)$ then the Cauchy–Schwartz inequality implies that

$$(x(t), x^0)_0 = (x(t), x^1)_0 = 0\,. \tag{2.10}$$

Hence it follows that $t = \frac{1}{2}$ and $(x^0, x^1)_0 = -1$. Consequently, if we multiply x^0 by a minus one, if necessary, we can insure that $(x^0, x^1)_0 \geqslant 0$ so that $\|x(t)\|_0 \neq 0$ for $t \in [0,1]$.

In the next two cases we assume $\langle x^0, x^1\rangle_0 \neq 0$.

Case 2. We multiply either x^0 or x^1 by a minus one, if necessary, to guarantee that

$$-\operatorname{sgn} \kappa (u, x^0)_0 \geqslant 0, \quad (u, x^1)_0 \geqslant 0\,. \tag{2.11}$$

Also, we choose any $\lambda > 0$ such that $\lambda(x^0, x^1)_0 \neq \operatorname{sgn} \kappa$ and replace x^0 by λx^0 so that we also have

$$(x^0, x^1)_0 \neq \operatorname{sgn} \kappa\,, \tag{2.12}$$

without changing the inequalities (2.11). With these conditions we observe that for (2.7) $\|x(t)\|_0 \neq 0$ for $t \in [0,1]$. If, on the contrary there is a $t \in (0,1)$ where $\|x(t)\|_0 = 0$, then it follows that $(x(t), u)_0 = 0$. Therefore the inequalities (2.11) imply

$$(u, x^0)_0 = (u, x^1)_0 = (u, u)_0 = 0\,.$$

Also, we obtain from the equation $(x(t), x^0)_0 = (x(t), x^1)_0 = 0$ that

$$\begin{aligned} (1-t)^2 + t^2 (x^0, x^1)_0 &= 0\,, \\ (1-t)^2 (x^0, x^1)_0 + t^2 &= 0\,, \end{aligned} \tag{2.13}$$

if $\kappa < 0$, while for $\kappa > 0$ we get

$$\begin{aligned} -(1-t)^2 + t^2 (x^0, x^1)_0 &= 0\,, \\ -(1-t)^2 (x^0, x^1)_0 + t^2 &= 0\,. \end{aligned}$$

Consequently, we conclude that $t = \frac{1}{2}$ and

$$(x^0, x^1)_0 = \operatorname{sgn} \kappa\,. \tag{2.14}$$

This last equation contradicts (2.12) and proves the theorem in this case.

Case 3. In this case, for $z \in \mathcal{H}$, $\langle x^0, z\rangle = \langle x^1, z\rangle = 0$ implies $\langle z, z\rangle = 0$. If there is a z^0 such that $\langle x^0, z^0\rangle = \langle x^1, z^0\rangle = 0$ and $(z^0, z^0)_0 \neq 0$ then the path (2.9) can be arranged so that $\|x(t)\|_0 \neq 0$ by choosing x^0, x^1, so that

$$(z^0, x^1)_0 \geqslant 0, \quad (z^0, x^0)_0 \geqslant 0 .$$

To see this we assume to the contrary that $\|x(t)\|_0 = 0$ for some $t \in (0, 1)$. For $t \in (0, \frac{1}{2}]$ we get from the equations $(x(t), x^0)_0 = (x(t), z^0)_0 = 0$ that $t = \frac{1}{2}$ and $(z^0, x^0)_0 = (z^0, z^0)_0 = 0$. Similarly, if $t \in [\frac{1}{2}, 1)$ we still conclude $(z^0, z^0)_0 = 0$. In either case this contradicts our choice of z^0. Thus the only possibility that remains is that $\langle x^0, z\rangle = \langle x^1, z\rangle = 0$ always implies $(z, z)_0 =$, i.e. $z \in \mathcal{N}$. Hence we see that *every* $z \in \mathcal{H}$ can be written as

$$z = \frac{\langle z, x^1\rangle}{\langle x^0, x^1\rangle} x^0 + \frac{\langle z, x^0\rangle}{\langle x^0, x^1\rangle} x^1 + u ,$$

where $u \in \mathcal{N}$. That is, $\mathcal{H} = \mathcal{S} + \mathcal{N}$, where $\mathcal{S} = \operatorname{span}\{x^0, x^1\}$. This contradicts our hypothesis that $\dim \mathcal{H} \geqslant 3$ and thus proves the theorem. □

Remark 2.1

It is essential in theorem 2.1 that $\dim \mathcal{H} \geqslant 3$, otherwise $\mathcal{M}$ is not convex. For two dimensional linear spaces $\mathcal{M}$ is typically an arc of a curve. In [5] we pointed out that generally $\mathcal{M}$ has a *convex* profile. This means that the boundary of the convex hull of $\mathcal{M}$ is contained in $\mathcal{M}$.

For the convenience of the reader we prove here that $\mathcal{M}$ always has a convex profile.

PROPOSITION 2.2

Let $\| \bullet \|_i$, $i = 0, 1, 2$, be quadratic semi-norms defined on some linear space $\mathcal{H}$. Then $\mathcal{M}$ has a convex profile.

Proof

We let $a(x) = (\|x\|_1^2, \|x\|_2^2)$ and suppose that $a(x^i) \in \mathcal{M}$, $i = 0, 1$. We multiply x^0 by minus one, (if necessary) so that $(x^0, x^1)_0 \geqslant 0$. Then $\|tx^0 + (1-t)x^1\|_0 > 0$, for *all* $t \in \mathbb{R}$. Therefore

$$x(t) = a(w(t)) ,$$
$$w(t) := (tx^0 + (1-t)x^1)/\|tx^0 + (1-t)x^1\|_0 ,$$

$t \in \mathbb{R}$ is a continuous curve lying in $\mathcal{M}$ connecting x^0 to x^1. Suppose now $\{a : \lambda \bullet a \geqslant \rho, a \in \mathbb{R}^2\}$, $\rho \in \mathbb{R}^1$, $\lambda \in \mathbb{R}^2$ is a half-space containing $\mathcal{M}$ and touching it at $a(x^0)$ and $a(x^1)$. Then the quadratic polynomial

$$Q(t) = \|tx^0 + (1-t)x^1\|_0 (\lambda \bullet x(t) - \rho)$$

is non-negative for all t and vanishes for $t = 0$ or 1. Hence $\lambda \bullet x(t) = \rho$ for all t and therefore $x(t)$, $t \in [0, 1]$ is the line segment joining x^0 to x^1 which must lie in $\mathcal{M}$. □

3. Optimal estimation in a Hilbert space from inaccurate data

This section follows [5] closely and shows how the convexity of the set $\mathcal{M}$ leads to certain conclusions concerning the optimal recovery in Hilbert spaces. To this end, we begin with a linear operator U from a linear space X into another linear space Z. We suppose both spaces are endowed with Hilbert space semi-norms $\| \bullet \|_X$ and $\| \bullet \|_Z$, respectively. Our interest is to estimate Ux optimally from incomplete information on $x \in X$ which takes two forms. First, we assume $\|x\|_X \leqslant 1$. Typically, this bound reflects some smoothness constraint on x and the desire to bound the error of recovery of Ux in terms of $\|x\|_X$. Secondly, we have available an information operator I from X into another linear space Y. Ix should be thought of as certain computable information about x. For instance, if x is a function Ix could represent the value of x at several points in its domain. We suppose $\| \bullet \|_Y$ is also a Hilbert space semi-norm on Y and that we are given a value $y \in Y$ which is approximately Ix in the sense that for some $\epsilon > 0$ we have $\|Ix - y\|_Y \leqslant \epsilon$. An algorithm is *any* mapping $A : Y \to Z$ and Ay provides an estimate for Ux where all we know is that $\|x\|_X \leqslant 1$ and $\|Ix - y\|_Y \leqslant \epsilon$. Thus the algorithm A can have an error as large as

$$E(A) := \sup\{\|Ux - Ay\|_Z : \|x\|_X \leqslant 1, \|Ix - y\|_Y \leqslant \epsilon\} \tag{3.1}$$

and *no* algorithm can have an error smaller than

$$E = \inf\{E(A) : A : Y \to Z\}. \tag{3.2}$$

An optimal algorithm is one that achieves this infimum but generally one does not exist and even if it does, it may *not* be *linear* [5]. However, in the present situation when $\| \bullet \|_X$, $\| \bullet \|_Y$ and $\| \bullet \|_Z$ are Hilbert space semi-norms an *optimal linear* algorithm exists under very mild additional assumptions. To explain what we have in mind, we introduce the space $V = X \times Y$ and for every $\mu \in [0, 1]$ the Hilbert space semi-norm $\| \bullet \|_\mu^2$ on V given by

$$\|(x, y)\|_\mu^2 := (1 - \mu)\epsilon^{-2}\|y\|_Y^2 + \mu\|x\|_X^2, \tag{3.3}$$

where $(x, y) \in V$. The graph of I defined by

$$\mathcal{G}_I = \{(x, Ix) : x \in X\} \tag{3.4}$$

is a subset of V and for each $y \in Y$ the quantity

$$\inf\{\|(x, Ix - y)\|_\mu^2 : x \in X\} \tag{3.5}$$

represents the distance in the norm $\|\bullet\|_\mu$ of $(0,y)\in V$ to $\mathcal{G}_I$. In concrete examples the solution of (3.5) represents a "regularized" solution to the problem $Ix = y + e$, $\|e\|_Y \leqslant \epsilon$, relative to $\|\bullet\|_X$. In the special case when x is a smooth function, Ix corresponds to values of x at certain points in its domain, and $\|\bullet\|_X$ represents the L^2-norm of some derivative of x the solution of the extremal problem (3.5) is a smoothing spline.

Suppose (3.5) has a solution $\bar{x} = \bar{x}(y)$ then it is a *linear* function of y and it is reasonable to recover Ux from the algorithm

$$A_\mu y := U\bar{x}(y)\,. \tag{3.6}$$

Let us estimate $E(A_\mu)$ and try to adjust μ to get the least error. For the upper bound we use the obvious fact that the intersection of the two "balls" $\|x\|_X \leqslant 1$, $\|Ix - y\|_Y \leqslant \epsilon$ is contained in the ball $\|(x, Ix - y)\|_\mu \leqslant 1$, and so trivially

$$E(A_\mu) \leqslant \sup\{\|U(x - \bar{x}(y)\|_Z : \|(x, Ix - y)\|_\mu \leqslant 1\}\,. \tag{3.7}$$

Choose any $x\in X, y\in Y$ with $\|(x, Ix - y)\|_\mu \leqslant 1$ set $u = x - \bar{x}(y)$ and observe that

$$\|(u, Iu)\|_\mu = \|(x, Ix) - (\bar{x}(y), I\bar{x}(y)\|_\mu\,.$$

Since $(\bar{x}(y), I\bar{x}(y))$ is the element of $\mathcal{G}_I$ which is closest to $(0,y)$ in the semi-Hilbert space $(V, \|\bullet\|_\mu)$ we have

$$\|(u, Iu)\|_\mu \leqslant \|(0,y) - (x, Ix)\|_\mu = \|(x, Ix - y)\|_\mu \leqslant 1\,.$$

Therefore (3.7) gives us the upper bound

$$E(A_\mu) \leqslant \sup\{\|Ux\|_Z : (1-\mu)\epsilon^{-2}\|Ix\|_Y^2 + \mu\|x\|_X^2 \leqslant 1\}\,. \tag{3.8}$$

For the lower bound we choose any algorithm A and any $x\in X$ such that $\|x\|_X \leqslant 1$ and $\|Ix\|_Y \leqslant \epsilon$. Then comparing Ix to $y = 0$ we get

$$E(A) \geqslant \|Ux - A0\|_Z$$

and if we replace x by $-x$ we also get

$$E(A) \geqslant \|Ux + A0\|_Z\,.$$

Thus the triangle inequality implies

$$E(A) \geqslant \sup\{\|Ux\|_Z : \|x\|_X \leqslant 1, \|Ix\|_Y \leqslant \epsilon\}\,. \tag{3.9}$$

We are going to use theorem 2.1 and adjust μ so that the upper bound in (3.8) matches the lower bound in (3.9). To this end, we define for $x\in X$

$$\|x\|_0 := \|Ux\|_Z, \quad \|x\|_1 = \epsilon^{-1}\|Ix\|_Y\,,$$

and

$$\|x\|_2 = \|x\|_X\,.$$

We claim that

$$\min\{\sup\{\|x\|_0 : (1-\mu)\|x\|_1^2 + \mu\|x\|_2^2 \leqslant 1\} : 0 \leqslant \mu \leqslant 1\}$$
$$= \sup\{\|x\|_0 : \max(\|x\|_1, \|x\|_2) \leqslant 1\}. \tag{3.10}$$

Equivalently, (3.10) says that

$$\inf_{a\in\mathcal{M}} \|a\|_\infty = \max_{\|b\|_1=1} \inf_{a\in\mathcal{M}} b \bullet a, \tag{3.11}$$

where $\mathcal{M}$ is given by (2.1), $\|\bullet\|_\infty$, $\|\bullet\|_1$ are the l^∞, l^1 on $\mathbb{R}^2$ respectively, and $b \bullet a$ is the inner product of the vectors a and b. For the equivalence of (3.11) and (3.10) we used the fact that $\mathcal{M} \subseteq \mathbb{R}_+^2$ and so the vector b which achieves the maximum in (3.11) has nonnegative components and so $b = (1-\mu, \mu)$ for some $\mu \in [0, 1]$. Now, any *convex* set $\mathcal{M}$ in $\mathbb{R}^2$ with $0 \notin$ interior $\mathcal{M}$ satisfies (3.11) and so in particular we conclude there is a $\mu \in [0, 1]$ such that

$$E(A_\mu) = E = \sup\{\|Ux\|_Z : \|x\|_X \leqslant 1, \|Ix\|_Y \leqslant \epsilon\}$$

when $\dim\mathcal{M} \geqslant 3$. This condition on $\mathcal{M}$ can be removed by using the fact that $\mathcal{M}$ has a convex profile because in this case (3.11) remains valid, see [5].

4. Optimal estimation in a Hilbert space from a finite number of inaccurate moments

We now consider the recovery of a vector x in a Hilbert space $\mathcal{H} = (\mathcal{H}, \|\bullet\|)$ from a finite number of its "moments" which are available with error

$$\sum_{k=0}^{N} |(x, \varphi_k) - u_k|^2 \leqslant \epsilon \tag{4.1}$$

and some a priori quadratic bound on x

$$\|x\|_X \leqslant 1. \tag{4.2}$$

We estimate the error in recovery with the norm on $\mathcal{H}$ and use the standard l_2^{N+1} norm on $\mathbb{R}^{N+1}$. Specifically, we are interested in determining

$$E_N(\mathcal{S}_N) := \inf_A \sup\{\|x - A\mu\| : \|x\|_X \leqslant 1, \|I_n x - \mu\|_N \leqslant \epsilon\} \tag{4.3}$$

where $\mathcal{S}_N := \operatorname{span}\{\varphi_0, \ldots, \varphi_N\}$,

$$I_N x := ((x, \varphi_0), \ldots, (x, \varphi_N)),$$

$$\|y\|_N^2 := \sum_{j=0}^{N} |y_j|^2, \quad y = (y_0, \ldots, y_N) \in \mathbb{R}^{N+1},$$

and A is any mapping from $\mathbb{R}^{N+1}$ into $\mathcal{H}$. The special case, $\varphi_j(x) = x^j, j = 0, 1, \ldots, N, \mathcal{H} = L^2[0, 1]$, so that

$$\|x\|^2 = \int_0^1 |x(t)|^2 \mathrm{d}t$$

and

$$\|x\|_X^2 := \int_0^1 |x^{(k)}(t)|^2 \mathrm{d}t,$$

where $x \in W_2^k[0, 1] := \{f : f, \ldots, f^{(k-1)}$ are absolutely continuous on $[0, 1], f^{(k)} \in L^2[0, 1]\}$ was studied in [10], when $k = 1$ however, the exact value of E_N or an optimal algorithm was not identified. Before we address this issue we generalize some observations made in [10]. For this purpose, we need the following lemma.

LEMMA 4.1

Let $\mathcal{H} = (\mathcal{H}, \| \bullet \|)$ be a Hilbert space and suppose $\mathcal{S}_N := \operatorname{span}\{\varphi_0, \ldots \varphi_N\}$ is an $N + 1$ dimensional subspace of $\mathcal{H}$. Let $\rho = \rho(\mathcal{S}_N) > 0$ be the largest constant such that

$$\rho^2 \sum_{j=0}^{N} |c_j|^2 \leqslant \left\| \sum_{j=0}^{N} c_j \varphi_j \right\|^2, \tag{4.4}$$

for all $c = (c_0, \ldots, c_N) \in \mathbb{R}^{N+1}$. Let $P_N : \mathcal{H} \to \mathcal{S}_N$ be the orthogonal projection of $\mathcal{H}$ onto $\mathcal{S}_N$, then

$$\|P_N x\|^2 \leqslant \rho^{-2} \sum_{j=0}^{N} |(x, \varphi_j)|^2. \tag{4.5}$$

Proof

For every $x \in \mathcal{H}$ there exist constants $c_0, \ldots, c_N$ such that $Px = \sum_{j=0}^{N} c_j \varphi_j$. Therefore

$$\begin{aligned} \|Px\|^2 = (Px, Px) = (x, Px) &= \sum_{j=0}^{N} c_j (x, \varphi_j) \\ &\leqslant \left(\sum_{j=0}^{N} |c_j|^2 \right)^{1/2} \left(\sum_{j=0}^{N} |(x, \varphi_j)|^2 \right)^{1/2} \\ &\leqslant \rho^{-1} \|Px\| \left(\sum_{j=0}^{N} (x, \varphi_j)|^2 \right)^{1/2}. \end{aligned}$$

□

Note that ρ in the inequality (4.4) is the square root of the least eigenvalue of the symmetric matrix (of moments)

$$M(\mathcal{S}_N) := \left(\int_0^1 \varphi_i(t)\varphi_j(t)\mathrm{d}t\right)_{i,j=0,1,\ldots,N}$$

When $\varphi_j(t) = t^j, 0 \leqslant j \leqslant N$, so that $\mathcal{S}_N = \pi_N$, all polynomials of degree $\leqslant N$, it was shown in [12] that, for some constant a, independent of N,

$$aN^{-1/2}b^N \geqslant \rho_N^{-1}, \tag{4.6}$$

where

$$b := (3 + 2\sqrt{2})^2.$$

Next, we consider the quantity

$$\tau(\mathcal{S}_N) := \sup\{\|x - P_N x\| : \|x\|_X \leqslant 1\}. \tag{4.7}$$

For instance, when $\mathcal{S}_N = \pi_N$, and

$$\|x\|_X^2 := \int_0^1 |x^{(k)}(t)|^2 \mathrm{d}t,$$

where $k \leqslant N + 1$, we have

$$\|x - Px\|_2 \leqslant (N + 2 - k)^{-k} c_k \|x\|_X \tag{4.8}$$

for some constant c_k depending only on k. This is a special case of Jackson's theorem. It was shown in [10] that $c_1 \leqslant 2^{-1/2}$ and in general it is known that $c_k \leqslant 2^{-k+1/2}$.

To prove this fact, we recall some properties of ultraspherical polynomials from [9]. Specifically, we have that

$$p_n^{(k,k)}(x) = \frac{(-1)^n}{2^n n!}(1 - x^2)^{-k}\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^n ((1 - x^2)^{n+k}), \quad n = 0, 1, \ldots, \tag{4.9}$$

are orthogonal relative the weight function $(1 - x^2)^k \mathrm{d}x$ on $[-1, 1]$ and

$$\int_{-1}^1 (P_n^{(k,k)}(x))^2 (1 - x^2)^k \mathrm{d}x = \frac{2^{2k+1}}{2n + 2k + 1} \frac{(n + k)!^2}{n!(n + 2k)!}. \tag{4.10}$$

According to (4.9) we have for $n \geqslant k$ that

$$\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^k ((1 - x^2)^k P_{n-k}^{(k,k)}(x)) = \frac{(-1)^k 2^k n!}{(n - k)!} P_n^{(0,0)}(x). \tag{4.11}$$

Thus we have for $n \geqslant k$

$$\int_{-1}^1 x^{(k)}(t)(1 - t^2)^k P_{n-k}^{(k,k)}(t)\mathrm{d}t = \frac{2^k n!}{(n - k)!} \int_{-1}^1 x(t) P_n^{(0,0)}(t)\mathrm{d}t. \tag{4.12}$$

Call the integral on the right-hand side of this equation λ_n. Then from (4.10) with $k = 0$ we get

$$\int_{-1}^{1} |x(t) - (P_N x)(t)|^2 \mathrm{d}t = \sum_{n=N+1}^{\infty} \tfrac{1}{2}(2n+1)\lambda_n^2 .$$

On the other hand, for the same reason (4.10) and (4.12) give

$$\begin{aligned}\int_{-1}^{1} (x^{(k)}(t))^2 (1-t^2)^k \mathrm{d}t &= \sum_{n=0}^{\infty} \tfrac{1}{2}(2n+2k+1)\frac{(n+2k)!}{n!}\lambda_{n+k}^2 \\ &= \sum_{n=k}^{\infty} \tfrac{1}{2}(2n+1)\frac{(n+k)!}{(n-k)!}\lambda_n^2 \\ &\geqslant \sum_{n=N+1}^{\infty} \tfrac{1}{2}(2n+1)\frac{(n+k)!}{(n-k)!}\lambda_n^2 \\ &\geqslant (N+2-k)^{2k} \int_{-1}^{1} |x(t) - (P_N x)(t)|^2 \mathrm{d}t .\end{aligned}$$

Thus we get

$$\int_{-1}^{1} |x(t) - P_N(t)|^2 \mathrm{d}t \leqslant (N+2-k)^{-2k} \int_{-1}^{1} (x^{(k)}(t))^2 \mathrm{d}t$$

and so changing variables of integration to the interval [0, 1] we get in (4.8) that $c_k \leqslant 2^{-k+1/2}$, as claimed. □

PROPOSITION 4.1

Suppose $\dim \mathcal{S}_n = n+1$ for $0 \leqslant n \leqslant N$. Then

$$E(\mathcal{S}_N) \leqslant \min\{\epsilon\rho(\mathcal{S}_n)^{-1} + \tau(\mathcal{S}_n) : 0 \leqslant n \leqslant N\}$$

and in particular

$$E(\pi_N) \leqslant \min\{\epsilon a n^{-1/2} b^n + (n+2-k)^{-k} : 0 \leqslant n \leqslant N\} .$$

Proof

Fix any n between zero and N, then

$$E(\mathcal{S}_n) \leqslant \sup\{\|x - P_n y\| : \|x\|_X \leqslant 1, \|I_n x - I_n y\|_n \leqslant \epsilon\} . \tag{4.13}$$

To see this, we first observe that there is an $A_n : \mathbb{R}^{n+1} \to \mathcal{H}$ such that

$$P_n x = A_n I_n x .$$

Thus,

$$E(\mathcal{S}_n) \leqslant \sup\{\|x - A_n \mu\| : \|x\|_X \leqslant 1, \|I_n x - \mu\|_n \leqslant \epsilon\}$$

and since I_n is onto we have

$$= \sup\{\|x - A_n I_n y\| : \|x\|_X \leqslant 1, \|I_n x - I_n y\|_n \leqslant \epsilon\}$$
$$= \sup\{\|x - P_n y\| : \|x\|_X \leqslant 1, \|I_n x - I_n y\|_n \leqslant \epsilon\}.$$

According to lemma 4.1 we have

$$\|P_n(x-y)\| \leqslant \rho(\mathcal{S}_n)^{-1}\|Ix - Iy\|_n,$$

and so combining this inequality with (4.15) gives

$$E(\mathcal{S}_n) \leqslant \sup\{\|x - P_n x\| : \|x\|_X \leqslant 1\} + \epsilon \sup\{\|P_n y\| : \|Iy\|_n \leqslant 1\}$$
$$\leqslant \tau(\mathcal{S}_n) + \epsilon\rho(\mathcal{S}_n)^{-1}. \qquad \square$$

We end this section with some brief comments about the optimal algorithm for the solution of the finite moment problem. Our remarks are only in the way of specializing the results of section 3 to π_N and $W_2^k[0,1]$ to obtain

$$E(\pi_N) = \max\{\|x\|_2 : \|x\|_{k,2} \leqslant 1, \|I_N\|_N \leqslant \epsilon\}$$
$$= \min_{0<\lambda<1} \max\{\|x\|_2 : (1-\lambda)\|x\|_{k,2}^2 + \lambda\epsilon^{-2}\|I_N x\|_N^2 \leqslant 1\}.$$

Let $\delta \in (0,1)$ achieve the minimum above. For any $\mu = (\mu_0, \ldots, \mu_N)$ let $x_{\text{opt}}(\bullet, \mu)$ be the solution of the variational problem

$$\min_{x \in W_2^k[0,1]} \left\{ (1-\delta)\int_0^1 |x^{(k)}(t)|^2 \mathrm{d}t + \delta\epsilon^{-2}\sum_{j=0}^{N}\left|\mu_j - \int_0^1 t^j x(t)\mathrm{d}t\right|^2 \right\}.$$

Then $A(\mu) = x_{\text{opt}}(\bullet, \mu)$ is an optimal linear algorithm.

5. Optimal information under uncertainty

In this section, we explore the question of choosing among all N-dimensional information operators $I_N x = ((x, \varphi_1), \ldots, (x, \varphi_N))$ the one that yields the smallest intrinsic error in optimal recovery. We assume throughout that X, Y, Z are Hilbert spaces of finite dimension with $Y = \mathbb{R}^N$. For a given information operator I_N the intrinsic error is given by

$$E(I_N) = \sup\{\|Ux\|_Z : \|x\|_X \leqslant 1, \|I_N x\|_Y \leqslant \epsilon\}.$$

We wish to identify the least value of $E(I_N)$ over a specified class of N-dimensional information operators. To this end, we let $\mathcal{I}_N$ be a given class of information operators with range in $\mathbb{R}^N$ and define

$$d_N(\epsilon) := \inf\{E(I_N) : I_N \in \mathcal{I}_N\}.$$

The positive semi-definite symmetric matrix $U^{\mathrm{T}}U : X \to X$, where U^{T} = transpose of U has eigenvalues $0 \leqslant \lambda_m \leqslant \ldots \leqslant \lambda_0$ and corresponding orthonormal eigenvectors $w^0, \ldots, w^m \in X$,

$$U^{\mathrm{T}}Uw^j = \lambda_j w j, \quad j = 0, \ldots, m.$$

We always assume $\mathfrak{I}_N$ contains the information operator

$$I_N x = ((x, w^0), \ldots, (x, w^{N-1})).$$

THEOREM 5.1

(i) If $I \in \mathfrak{I}_N$ implies that $\lambda I \in \mathfrak{I}_N$ for all $\lambda > 0$ then

$$d_N^2(\epsilon) = \begin{cases} \lambda_N, & N \leqslant m, \\ 0, & N < m. \end{cases} \tag{5.1}$$

(ii) If $I \in \mathfrak{I}_N$ implies that $\|Iw^0\| \leqslant 1$ then

$$d_N^2(\epsilon) = \begin{cases} \epsilon^2\lambda_0 + (1-\epsilon^2)\lambda_N, & N \leqslant m, 0 \leqslant \epsilon \leqslant 1, \\ \lambda_0, & \epsilon \geqslant 1. \end{cases} \tag{5.2}$$

Proof

We first consider (i). In this case it is obvious that $d_N(\epsilon)$ is independent of ϵ and so

$$\begin{aligned} d_N(\epsilon) &= d_N(0) \\ &= \sup\{\|Ux\|_Z : \|x\|_X \leqslant 1, I_N x = 0\}. \end{aligned}$$

Thus (5.1) follows from the well-known min-max characterization of eigenvalues.

For (ii) we first consider the case that $N \leqslant m$ and $\epsilon \in [0, 1]$. Given any information operator

$$Ix = ((x, \varphi_1), \ldots, (x, \varphi_N)),$$

in $\mathfrak{I}_N$ we set

$$x = \epsilon w^0 + \sqrt{1-\epsilon^2} \sum_{j=0}^{N} c_j w^j$$

and choose $(c_0, \ldots, c_N)$ such that

$$I\left(\sum_{j=0}^{N} c_j w^j\right) = 0, \quad \sum_{j=0}^{N} c_j^2 = 1, \quad c_0 \geqslant 0.$$

Then

$$\|x\|_X^2 = 1 + 2\epsilon\sqrt{1-\epsilon^2}c_0,$$

so that $\|x\|_X \geqslant 1$ and $Ix = \epsilon I w^0$ which implies that $\|Ix\|_Y \leqslant \epsilon$. Moreover, we have

$$\begin{aligned}
\|Ux\|_Z^2 &= (U^{\mathrm{T}}Ux, x) \\
&= \lambda_0(\epsilon + \sqrt{1-\epsilon^2}c_0)^2 + (1-\epsilon^2)\sum_{j=1}^{N}\lambda_j c_j^2 \\
&\geqslant \lambda_0(\epsilon + \sqrt{1-\epsilon^2}c_0)^2 + \lambda_N(1-c_0^2)(1-\epsilon^2) \\
&= \lambda_0\epsilon^2 + (1-\epsilon^2)\lambda_N + 2\lambda_0\epsilon\sqrt{1-\epsilon^2}c_0 \\
&\quad + (1-\epsilon^2)(\lambda_0 - \lambda_N)c_0^2 \\
&\geqslant (\lambda_0\epsilon^2 + (1-\epsilon^2)\lambda_N)\{1 + 2\epsilon\sqrt{1-\epsilon^2}c_0\} \\
&= (\lambda_0\epsilon^2 + (1-\epsilon^2)\lambda_N)\|x\|_X^2 .
\end{aligned}$$

Therefore we obtain the lower bound

$$d_N^2(\epsilon) \geqslant \lambda_0\epsilon^2 + (1-\epsilon^2)\lambda_N .$$

For the upper bound on $d_N^2(\epsilon)$ we choose

$$I_N x = ((x, w^0), \ldots, (x, w^{N-1}))$$

so that

$$\begin{aligned}
&\max\{\|Ux\|_Z^2 : \|x\|_X \leqslant 1, \|I_N x\|_Y \leqslant \epsilon\} \\
&\quad = \max\left\{\sum_{l=0}^{m}\lambda_l c_l^2 : \sum_{l=0}^{m} c_l^2 \leqslant 1, \sum_{l=0}^{N-1} c_l^2 \leqslant \epsilon^2\right\} .
\end{aligned}$$

By (3.10) this equals

$$\begin{aligned}
&\min_{0\leqslant\tau\leqslant 1}\max\left\{\sum_{l=0}^{m}\lambda_l c_l^2 : (\epsilon^{-2}(1-\tau)+\tau)\sum_{l=0}^{N-1} c_l^2 + \tau\sum_{l=N}^{m} c_l^2\right\} \\
&\quad = \min_{0\leqslant\tau\leqslant 1}\max\left(\frac{\lambda_0}{(1-\tau)\epsilon^{-2}+\tau}, \frac{\lambda_N}{\tau}\right) \\
&\quad = \epsilon^2\lambda_0 + (1-\epsilon^2)\lambda_N .
\end{aligned}$$

Therefore

$$\max\{\|Ux\|_Z^2 : \|x\|_X \leqslant 1, \|I_N x\|_Y \leqslant \epsilon\} = \epsilon^2\lambda_0 + (1-\epsilon^2)\lambda_N ,$$

which proves the first part of (ii).

For the second part of (ii) we see from the argument above that

$$\max\{\|Ux\|_Z^2 : \|x\|_X \leqslant 1, \|I_N x\|_Y \leqslant \epsilon\} \leqslant \lambda_0 .$$

Also, if $I \in \mathfrak{I}_N$ then $\|Iw^0\|_Y \leqslant \epsilon$, $\|w^0\|_X = 1$ and $\|Uw^0\|_Z^2 = \lambda_0$ and so $d_N^2(\epsilon) \geqslant \lambda_0$. □

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